CCA Exam Management System

System Demonstration Guide

Prepared for Panel Presentation

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System URL: http://127.0.0.1:9999

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1. System Overview

The **CCA Exam Management System** is a full-stack web application built to automate and streamline the exam scheduling, proctoring, and management process. It replaces manual spreadsheet-based scheduling with an intelligent algorithm that respects real-world constraints.

Technology Stack

Layer	Technology
Backend	Python 3 / Flask
Database	MySQL (via XAMPP)
Frontend	HTML5, TailwindCSS, JavaScript
Authentication	bcrypt password hashing
Email	Flask-Mail (SMTP integration)
Export	ReportLab PDF generation
Architecture	MVC pattern with modular system/ package

Key System Files

File	Purpose
app.py	Main Flask application — all routes (1,277 lines)
system/database.py	Database access layer with 50+ functions
system/scheduler.py	Automated scheduling algorithm (279 lines)
system/exporter.py	PDF export functionality
system/logic.py	Authentication & password hashing
system/importer.py	CSV import parsing

2. Authentication & Role-Based Access

The system uses **session-based authentication** with bcrypt-hashed passwords. Upon login, users are automatically routed to their role-specific dashboard.

Login Page

- Clean login form with email and password fields
- Flash message support for errors and success notifications
- Link to student self-registration

Five User Roles

Role	Dashboard	Access Level
Super Admin	/super_admin	Full system control, user management
Admin	/dashboard	Exam/room/instructor CRUD, scheduling
Proctor	/proctor	View assignments, set availability
Secretary	/secretary	Track exam paper collection/return
Student	/timetable	View personalized exam timetable

Student Registration

- Students select their **institute** and **section** during registration
- Passwords are **bcrypt-hashed** before storage
- Duplicate email addresses are automatically rejected

3. Admin Dashboard — The Control Center

The admin dashboard at **/dashboard** serves as the central hub for managing all exam-related data. It provides real-time statistics and quick-access forms organized into tabbed sections.

Room Management

- Add rooms with: name, capacity, building, floor, institute
- **PWD accessibility flag** — ensures wheelchair-accessible rooms are properly tracked
- **Bulk CSV import** — upload rooms in batch via /import/rooms

Instructor / Faculty Management

- Add instructors with: name, department, institute, building, PWD flag
- **Subjects taught** field — prevents conflict-of-interest proctoring
- Activate/deactivate toggle — only active instructors are used during scheduling
- When an instructor is added, a **proctor login account** is automatically created
- **Bulk CSV import** — upload instructors via /import/instructors

Exam Management

- Create exams with: subject name, section, type (Major/Minor), duration, institute
- Exams start with **Pending** status until scheduled
- **Bulk delete** — admin can clear all exams for fresh scheduling

4. Institute & Subject Management

Institutes

Academic institutes (e.g., ICSLIS, IBED, ITHM) are the organizational units of the system. Each institute has a unique **code** and a **default capacity**. Institutes are linked to rooms, instructors, subjects, and exams.

Subjects

- Full CRUD operations with institute association
- **Common Subject flag** — marks subjects taken across all institutes (e.g., MATH 2, FILIPINO 2, ENG 2)
- **Content Group Linking** — subjects with **different names** but the **same exam content** can be linked so the scheduler assigns them the same time slot
- **CSV Import** — bulk upload subjects with institute mapping

The Problem: Same Subject, Different Names

Real-world issue at CCA: Some subjects are offered across different programs and institutes but have **different subject codes or names** — even though the exam content is identical. Without proper handling, these exams could be scheduled at different times, allowing students to share answers between time slots.

Real CCA Examples

Example 1: Subjects with the SAME name across DIFFERENT programs

These subjects appear under the same code but are taken by students from different programs. The system automatically groups them into the same time slot:

Subject Code	Sections	Programs	Scheduled Together?
MATH 2	BPE 202, BSNE 201	Phys. Ed. & Secondary Ed.	YES — same time slot
EXPTPSY	PSYCH 201–205 (5 sections)	Psychology	YES — all at 8:00 AM
IBT	E401–E405 (5 sections)	Education	YES — all at 8:00 AM
TOURCULDIV	T301–T307 (7 sections)	Tourism & Hospitality	YES — all at 8:00 AM
INTROCOUN	PSYCH 201–205 (5 sections)	Psychology	YES — all at 9:00 AM
PE 108	BPE 201, BPE 202	Physical Education	YES — all at 9:00 AM

Example 2: Common (GE) subjects across ALL institutes

General Education subjects are taken by students from all institutes. These are flagged as **Common Subjects** in the system:

Subject	Code	Appears In	Common Flag
Mathematics 2	MATH 2	BPE, BSNE (and potentially other programs)	YES
Filipino 2	FILIPINO 2	All institutes — General Education	YES
English 2	ENG 2	All institutes — General Education	YES

Example 3: Subjects with DIFFERENT names but SAME content (Content Group Linking)

This is where Content Group Linking becomes critical. In CCA, some institutes rename or recode the same subject differently:

Institute A	Subject Name A	Institute B	Subject Name B	Action Required
General Ed.	MATH 2	Engineering	GE-MATH 2	Link as Content Group
ICSLIS	8MAEL1	Education	Applied Linguistics	Link as Content Group
Tourism	TOURCULDIV	Hospitality	Cultural Studies	Link as Content Group

How Content Group Linking Works

- 1. Navigate to the **Subjects** page (/subjects)
- 2. Find the **"Link Identical Content"** form
- 3. Select **Subject A** (e.g., MATH 2 from General Ed.)
- 4. Select **Subject B** (e.g., GE-MATH 2 from Engineering)
- 5. Click **Link** — both subjects now share a content_group_id
- 6. When the scheduler runs, it groups them and assigns the **same time slot**

Result: Even though the subjects have different names in the system, all linked exams are scheduled simultaneously — making it **impossible for students to leak exam content** across time slots. This is one of the strongest features of the system.

Visual Summary

WITHOUT Content Group Linking	WITH Content Group Linking
MATH 2 (BPE) → 8:00 AM GE-MATH 2 (Eng.) → 10:00 AM ■ **RISK:** Students can share answers between 8 AM and 10 AM slots	MATH 2 (BPE) → 8:00 AM GE-MATH 2 (Eng.) → 8:00 AM ✓ **SAFE:** All identical exams happen at the same time — no leaks possible

5. The Automated Scheduler (Core Algorithm)

This is the heart of the system. The scheduler at /schedule takes all Pending exams and automatically assigns them to time slots, rooms, and proctors while respecting 7+ real-world constraints.

How It Works

- 1. Admin sets the **exam period** (start and end dates)
- 2. Clicks **Generate Schedule**
- 3. The algorithm processes all Pending exams through the constraint engine
- 4. Results are saved and displayed as a timetable grid

Scheduling Constraints Enforced

The scheduler respects **7 constraint rules** simultaneously:

#	Constraint	How It Works
1	Max consecutive proctor hours	Tracks cumulative blocks per proctor per day (configurable, default 8 hrs)
2	No self-subject proctoring	A proctor will never supervise an exam in their own subject
3	Max 2 majors per section per day	Prevents student overload — tracked via maj_count[section][date]
4	Room capacity & availability	Time-interval overlap detection ensures no double-booking
5	Proctor availability windows	Honors each proctor's declared available dates and times
6	PWD proctor matched to PWD room	Proctors with disabilities are only assigned to accessible rooms
7	Travel buffer between buildings	Configurable gap (default 15 min) between assignments in different buildings

Content Group Scheduling

Subjects linked as **identical content** are grouped and scheduled at the **same time slot** across all sections — this is critical for preventing exam content leaks between time slots.

Algorithm Flow

- 1. Retrieve all Pending exams from the database
- 2. Group exams by content group or subject name
- 3. Sort groups: Major exams first, then by name
- 4. For each group, iterate through all date + slot combinations
- 5. Check all 7 constraints for each candidate slot
- 6. If all constraints pass, greedily assign rooms and proctors
- 7. Lock the time slot for this subject (content group consistency)

- 8. If no valid slot found, mark group as Failed/Unscheduled

Post-Scheduling Actions

- **Email notifications** are automatically sent to assigned proctors via SMTP
- **In-app notifications** are created for all affected users
- Failed exams are highlighted with **specific failure reasons**

6. Schedule View & Timetable

Admin Schedule View

The generated schedule is displayed as an **interactive timetable grid** where columns represent exam dates and rows represent time slots. Each cell shows the subject, section, room, and proctor.

Conflict Detection

- **Room double-booking** — same room assigned to overlapping times (highlighted in red)
- **Same subject at different times** — potential content leak risk (highlighted in yellow)

Inline Schedule Editing

- Admin can edit any scheduled exam: change room, proctor, or date/time
- **Real-time conflict checking** via AJAX call to /check_edit_conflict
- Force-save option to override warnings when necessary

Schedule Reset

The admin can reset the entire schedule, returning all exams to Pending status for re-scheduling.

Student Timetable (/timetable)

- Students see only **their section's and institute's exams**
- Exams grouped by date with clean cards showing time, subject, room, and proctor
- Read-only access — no editing capabilities

7. Proctor Dashboard

Accessible at **/proctor** by proctor-role users. This dashboard provides proctors with everything they need to manage their exam assignments.

Feature	Description
My Assignments	List of all exams they're proctoring, grouped by date with time and room
Total Hours	Running total of proctoring hours for the current exam period
Set Availability	Select available dates and time windows within the exam period
Report Absence	Self-report inability to proctor with reason — automatically notifies admin

8. Secretary Dashboard

The secretary dashboard at **/secretary** tracks exam paper logistics throughout the exam period.

Action	What It Does
Mark Collected	Records which instructor handed in exam papers, with optional notes
Mark Returned	Records when graded papers are returned to the department
Collection Stats	Live counters showing collected, returned, and pending papers

PDF Export: The secretary can download a collection report as a PDF document.

9. Reports & Analytics

Three Report Categories

1. Proctor Absence Report

- Filterable by date range
- Shows proctor name, exam, reason, and who reported the absence

2. Assignment Report

- Per-proctor breakdown of exam assignments
- Helps identify workload imbalance across faculty

3. Scheduling Report

- Overall statistics: total exams, scheduled vs. unscheduled
- Identifies scheduling bottlenecks and constraint conflicts

PDF Exports

Export	Route	Content
Exam Schedule	/export/schedule.pdf	Full timetable with conflict highlights
Absence Report	/export/absence_report.pdf	Proctor absence data + hours summary
Collection Report	/export/collection_report.pdf	Paper collection tracking status

10. Super Admin Panel

Accessible at **/super_admin**, restricted to users with the **is_super_admin** flag set.

Capabilities

- **View all users** across all roles in a single table
- **Edit any user:** change name, email, role, section, institute, and super admin flag
- **Delete users** with protection against self-deletion
- **Reset passwords** — generates a new bcrypt hash for the specified user

11. System Settings

The settings page at **/settings** allows admins to configure scheduling parameters and email integration.

Setting	Default	Purpose
Day Start	07:00	Earliest possible exam slot
Day End	20:00	Latest possible exam end time
Slot Interval	60 min	Time gap between consecutive slot start times

Setting	Default	Purpose
Max Proctor Hours	8 hours	Maximum consecutive proctoring hours per day
Travel Buffer	15 min	Required gap between assignments in different buildings
SMTP Configuration	—	Host, port, username, password, TLS settings for email

Test Email: Admins can send a test email to verify SMTP configuration is working correctly.

12. Notification System

- **In-app notifications** for all user roles
- **Unread badge counter** displayed on every page in the navigation
- **Mark read** — individually or all at once
- **Role-based notifications:** e.g., admin is notified when a proctor reports absence
- **Email notifications:** sent automatically for schedule assignments and changes via SMTP

13. Recommended Live Demo Flow (~18 minutes)

Act	Feature	Duration	Steps
1	Login & Routing	2 min	Open the system, show login page, log in as admin
2	Data Management	3 min	Show institutes, subjects, rooms tab, add a room, show instructors
3	Exam Creation	2 min	Add new exams, show Pending status
4	Automated Scheduling	3 min	Set exam period, generate schedule, show timetable grid, edit an exam
5	Proctor Experience	2 min	Log in as proctor, show assignments, set availability, report absence
6	Secretary Experience	1 min	Log in as secretary, mark collected/returned
7	Student Experience	1 min	Register a student, show personalized timetable
8	Reports & Export	2 min	Show all 3 report types, download PDF exports
9	Super Admin	1 min	Show user management and password reset
10	Settings	1 min	Show configurable parameters and SMTP config

Detailed Steps for Act 4 (Core Demo)

This is the most important part of the demonstration — the automated scheduling algorithm:

- 1. Navigate to **Schedule** page (/schedule)
- 2. Set the exam period dates (e.g., August 24-29, 2026)
- 3. Click **Generate Schedule** — watch the algorithm process all pending exams
- 4. Point out the generated timetable: room assignments, proctor assignments
- 5. Verify: no room conflicts, no self-subject proctoring, no overloaded sections
- 6. Click on a scheduled exam — demonstrate the **inline edit** feature
- 7. Change the room — show the **real-time conflict checker** in action
- 8. (Optional) Click **Reset Schedule** to return exams to Pending and re-run

14. Key Selling Points for Panel Q&A;

These are the differentiating features that set this system apart from manual or basic scheduling approaches:

#	Feature	Why It Matters
1	Constraint-Aware Scheduling	Not random assignment — respects 7+ real-world rules simultaneously
2	Content Group Linking	Prevents exam leaks by scheduling identical-content exams at the same time
3	PWD Accessibility	Matches PWD proctors with PWD-accessible rooms automatically
4	Travel Buffer Logic	Accounts for physical movement time between buildings
5	Multi-Role Architecture	5 distinct user roles with appropriate access levels and dashboards
6	Real-Time Conflict Detection	AJAX-powered checks when editing schedules — instant feedback
7	Full Audit Trail	Notifications, absence records, collection logs for accountability
8	PDF Exports	Professional reports for printing, archival, and distribution
9	Email Notifications	Automatic SMTP-based alerts for proctor assignments and changes
10	Bulk Data Import	CSV upload support for rooms, instructors, and subjects

To start the system: Ensure XAMPP/MySQL is running, then execute *python app.py* in the project directory. The application will be available at **http://127.0.0.1:9999**.